

UNIT 5

WORK, ENERGY & POWER



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WORK:

"Work is said to be done when a force displaced a body in the direction of force."

Formula:

Work is the dot product of force and displacement.

Work = (Force) (Displacement)

$$W = \vec{F} \cdot \vec{d}$$

If Force "F" makes an angle θ with the direction of motion:

$$W = F d \cos \theta$$

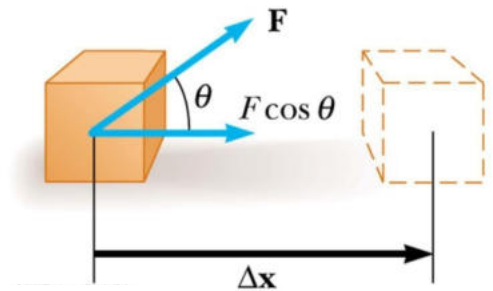
• Work is scalar quantity.

Units:

- In S.I system unit of work is Joule (J).
- In C.G.S system unit of work is Erg.
- In F.P.S system unit of work is Foot-Pound (ft - lb).
- 1 joule = 10^7 erg.

Dimension of work:

Dimension of work is ML^2T^{-2} .



TYPES OF WORK DONE:

• **Work done when the force and displacement are in same direction:**

• **Or Positive work done (When $\theta = 0^\circ$)/Maximum work done:**

If the force is in the same direction as the displacement, the work is positive. Here $\theta = 0^\circ$ the work will be positive and maximum.

$$W = F S \cos \theta$$

Where $\theta = 0^\circ$

$$W = F S \cos (0)$$

$$W = F S (1)$$

$$W = F S$$

Example:

If a spring is stretched the applied force and displacement is in the same direction so work done is positive.

Work done when the force and displacement are in opposite direction:

Or

Negative work done (When $\theta = 180^\circ$):

If the force is in the opposite direction as the displacement, the work is negative. Here $\theta = 180^\circ$ the work will be negative.

$$W = F S \cos \theta$$

Where $\theta = 180^\circ$

$$W = F S \cos (180)$$

$$W = F S (-1)$$

$$W = - F S$$

Example of negative W.D:

When the body is lifted upwards work done by gravity is negative.

Work done when the force is at right angle to the displacement:

Or

Minimum work done (When $\theta = 90^\circ$)/Zero work done:

If the force is at right angle to the displacement, the work is minimum. Here $\theta = 90^\circ$ the work will be zero minimum.

$$W = F S \cos \theta$$

$$\text{Where } \theta = 90^\circ$$

$$W = F S \cos (90)$$

$$W = F S (0)$$

$$W = 0$$

Example of zero W.D:

For a body moving in a circular path, the centripetal force and displacement are perpendicular to each other so work done is zero.

CONSERVATIVE FIELD:

"A field in which total work done along a closed path is zero is called conservative field."

Examples of conservative and non-conservative force:

NON-CONSERVATIVE FORCES	CONSERVATIVE FORCES
Static and kinetic frictional forces	Gravitational force
Air resistance	Elastic spring force
Tension	Electric force
Normal forces	
Propulsion force of a rocket	

WORK DONE BY GRAVITATIONAL FIELD IS INDEPENDENT OF PATH:

Work done in moving a body in gravitational field is independent of path and depends only the initial and final position of the body and for a closed path net work is zero.

Consideration:

Consider a rectangular closed path ABCA as shown in which a body of weight W is to be displaced from point A to B, B to C and finally from C to A in gravitational field. AD is perpendicular to the base BC in gravitational field.

Proof:

$$\begin{aligned} \text{Work}_{A \rightarrow B} &= \vec{F} \cdot \vec{d} \\ \text{Work}_{A \rightarrow B} &= W s_1 \cos \alpha \\ \text{Work}_{A \rightarrow B} &= mg s_1 \cos \alpha \dots\dots\dots (1) \end{aligned}$$

From triangle ABD,

$$\begin{aligned} \frac{\text{Base}}{\text{Hyp}} &= \frac{AD}{AB} = \cos \alpha \\ \frac{h}{s_1} &= \cos \alpha \end{aligned}$$

$$h = s_1 \cos \alpha$$

Substituting the value of $s_1 \cos \alpha$ in equation (1), we get:

$$\text{Work}_{A \rightarrow B} = mgh \dots\dots\dots (2)$$

Work done in moving the body along the path BC:

$$\begin{aligned} \text{Work}_{B \rightarrow C} &= \vec{F} \cdot \vec{d} \\ \text{Work}_{B \rightarrow C} &= W s_2 \cos \theta \\ \text{Work}_{B \rightarrow C} &= W s_2 \cos (90^\circ) \\ \text{Work}_{B \rightarrow C} &= 0 \dots\dots\dots (3) \end{aligned}$$

Work done in moving the body along the path CA:

$$\text{Work}_{C \rightarrow A} = \vec{F} \cdot \vec{d}$$

$$\text{Work}_{C \rightarrow A} = Fd \cos \theta$$

$$\text{Work}_{C \rightarrow A} = WS_3 \cos(180^\circ - \beta)$$

$$\text{Work}_{C \rightarrow A} = mgS_3(-\cos \beta) \quad \because \cos(180^\circ - \beta) = -\cos \beta$$

$$\text{Work}_{C \rightarrow A} = mg(-S_3 \cos \beta) \quad (3)$$

In triangle ACD,

$$\frac{h}{S_3} = \cos \beta$$

$$h = S_3 \cos \beta$$

Substituting the value of $S_3 \cos \beta$ in equation (4),

$$\text{Work}_{C \rightarrow A} = mg(-h)$$

$$\text{Work}_{C \rightarrow A} = -mgh$$

Total work done along closed path ABCA:

$$\text{Work}_{A \rightarrow B} + \text{Work}_{B \rightarrow C} + \text{Work}_{C \rightarrow A} = mgh + 0 + (-mgh)$$

$$\text{Work}_{A \rightarrow B} + \text{Work}_{B \rightarrow C} + \text{Work}_{C \rightarrow A} = mgh - mgh$$

$$\text{Work}_{A \rightarrow B} + \text{Work}_{B \rightarrow C} + \text{Work}_{C \rightarrow A} = 0$$

Conclusion:

Above equation shows that work done in close path in gravitational field is zero.

POWER:

"Rate of doing work is called power **OR** work done in unit time is called power."

Formula:

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

$$P = \frac{W}{t}$$

Units of power:

• In S.I system:

In S.I system unit of power is watt ($\text{J} \cdot \text{S}^{-1}$) after James Watt.

• In C.G.S system:

In C.G.S system unit of power is Erg per second ($\text{Erg} \cdot \text{S}^{-1}$).

• In British engineering system:

In B.E system unit of power is Horse Power (hp).

$$1 \text{ hp} = 746 \text{ Watts}$$

Dimension:

The dimension of power is $[\text{ML}^2\text{T}^{-3}]$.

POWER IN TERMS OF VELOCITY:

"The scalar product of force and velocity is called power."

Formula:

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

$$P = \frac{W}{t}$$

$$P = \frac{f \cdot d}{t} \because W = \vec{F} \cdot \vec{d}$$

$$P = f \cdot \frac{d}{t}$$

$$P = F \cdot V$$

RELATION BETWEEN WORK (ENERGY) AND POWER:

We know that,

$$\text{Power} = \frac{\text{Work}}{\text{Time}}$$

$$\text{Work} = \text{Power} \times \text{time}$$

$$\text{Work} = \text{Kilowatt} \times \text{hour}$$

$$\text{Energy} = \text{Kilowatt} \times \text{hour}$$

Thus the unit of energy is Kilowatt hour (Kwh).

Definition of Kilowatt hour:

"The amount of work done when power of 1 Kw is maintained for 1 hour is called 1 kilowatt hour."

Relation between kilowatt hour and joules:

- 1 kilowatt-hour = 1000 Watts × 1 hour
- 1 kilowatt-hour = 1000 J·Sec⁻¹ × 3600 Sec
- 1 kilowatt-hour = 3.6 × 10⁶ J

$$1 \text{ kilowatt-hour} = 3.6 \times 10^6 \text{ J}$$

ENERGY:

"The ability or capacity to do work is called energy."

Types of mechanical energy:

There are two types of mechanical energy:

- Kinetic energy.
- Potential energy.

KINETIC ENERGY:

"The energy possessed by a body by virtue of its motion is called kinetic energy."

Derivation for formula of kinetic energy:

Consider a body of mass 'm' which is projected up in the gravitational field with a velocity 'V'. After attaining a maximum height 'h' the body comes to rest. The initial kinetic energy of the body is capable of doing work and is used up in doing work against the force of gravity.

At maximum height the kinetic energy of the body is zero because it is no more capable of doing work against the gravitational force.

Work done by the body = F · d

$$W = Fd \cos \theta$$

$$W = Fh \cos \theta$$

$$W = mgh \cos \theta$$

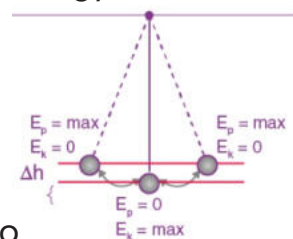
$$W = mgh$$

$$\because d = h$$

$$\because F = mg$$

$$\because \theta = 0$$

$$(i)$$



Eliminate 'h' from above equation by using equation of motion,

$$v^2 - v_i^2 = 2aS \dots\dots\dots (ii)$$

Where,

Initial velocity = $v_i = v$

Final velocity = $v_f = 0$

Acceleration = $a = -g$

Maximum height = $S = h$

Substituting these above values in equation (ii),

$$v^2 - v^2 = 2aS$$

$$(0)^2 - (v)^2 = 2(-g)h$$

$$-v^2 = -2gh$$

$$h = \frac{v^2}{2g} \quad (iii)$$

Substituting the value of 'h' from equation (iii) in equation (i) we get,

$$W = mgh$$

$$W = mg \cdot \frac{v^2}{2g}$$

$$K.E = \frac{1}{2} mv^2$$

This work is done due to kinetic energy therefore,

$$K.E = \frac{1}{2} mv^2$$

POTENTIAL ENERGY:

"Energy possessed by a body by virtue of its motion is called potential energy."

Different types of potential energy:

Depending upon the forces involved, the potential energy of the system can be different types.

1. Gravitational potential energy:

"The energy stored in a body when the work is done against the gravitational field is called gravitational potential energy."

2. Elastic potential energy:

"The energy stored in a body due to elastic restoring force is called elastic potential energy."

3. Electrostatic potential energy:

"The energy stored in a body due to electrostatic force is called elastic potential energy."

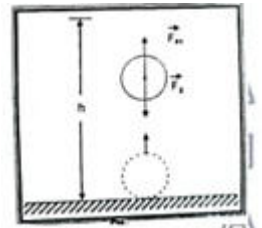
EXPRESSION FOR GRAVITATIONAL POTENTIAL ENERGY:

Consider a body of mass 'm' which is taken very slowly to small height 'h' in the gravitational field such that the acceleration of the body is zero. The work done in moving the body is given by:

$$\text{Work done} = F_{\text{ext}} \cdot r$$

$$\text{Work done} = F_{\text{ext}} \cdot h \cdot \cos\theta$$

Where F is the external force applied on the body. Since the external force applied on the body and the displacement are along the same direction, therefore work done by external force W_{ext} is given by:



$$W_{\text{ext}} = F_{\text{ext}} \cdot h \cdot \cos\theta$$

$$W_{\text{ext}} = F_{\text{ext}} \cdot h \cdot \cos 0^\circ = 1$$

As the acceleration of the body is zero therefore magnitude of external force is equal to that of the force of gravity i.e.,

$$F_{\text{ext}} = mg$$

$$W_{\text{ext}} = mgh \dots\dots\dots (i)$$

Work done " W_g " by gravitational force " F_g " is given by,

$$W_g = F_g \cdot r$$

$$W_g = mgh \cdot \cos\theta$$

Where gravitational force and displacement are in opposite direction therefore,

$$W_g = mgh \cdot \cos 180^\circ$$

$$W_g = -mgh \dots\dots\dots (i)$$

$$-W_g = mgh \dots\dots\dots (ii)$$

Comparing equation (i) and (iii),

$$W_{\text{ext}} = -W_g$$

By putting the value of W_g from equation (ii) we get,

$$W_{\text{ext}} = -W_g = -(-mgh) = mgh$$

This work done on the body is due to gravitational force, it is stored in the form of potential energy which is represented by " U_g ".

$$U_g = mgh$$

This gravitational potential energy is the relative potential energy of the body with respect to some arbitrary zero level.

ABSOLUTE GRAVITATIONAL POTENTIAL ENERGY:

"The work done in moving a body from earth's surface to zero point is called its absolute potential energy."

Zero point:

"Point which is so far away from the earth that an object, situated at it is not influenced by the earth's attraction is called zero point and potential energy at this point will be zero."

EXPRESSION FOR ABSOLUTE GRAVITATIONAL POTENTIAL ENERGY:

Consider a body of mass ' m ' at point A (0 or 1) in the gravitational field. If the body is lifted to a far off point B (or ' n ') in the gravitational field, then work done in moving body cannot be directly found by using the formula:

$$\text{Work} = (\text{Force}) \times (\text{Displacement})$$

Because the gravitational force will not remain constant for such a large distance.

To overcome the difficulty, divide the distance between A and B into a large number of intervals each of width Δr . ' Δr ' is so small that the gravitational force throughout this interval may be assumed to be constant.

If ' r ' be the average distance of r_1 and r_2 ,

$$F = G \frac{m_1 m_2}{r^2}, \quad (i)$$

Where r is the average distance of r_1 and r_2 :

$$r = \frac{r_1 + r_2}{2}$$

And,

$$r_2 - r_1 = \Delta r \Rightarrow r_2 = r_1 + \Delta r$$

So,

$$r = \frac{r_1 + r_1 + \Delta r}{2}$$

$$r = \frac{2r_1 + \Delta r}{2}$$

$$r^2 = \frac{4r_1^2 + 4r_1\Delta r + \Delta r^2}{4}$$

Δr is so small, so Δr^2 is negligible,

$$r^2 = \frac{4r_1^2 + 4r_1\Delta r}{4}$$

$$r^2 = r_1^2 + r_1\Delta r$$

$$r^2 = r_1^2 + r_1(r_2 - r_1)$$

$$r^2 = r_1 r_2$$

Putting r^2 in equation (i),

$$F = G \frac{m M_E}{r_1 r_2} \quad (\text{ii})$$

Work done in lifting the body from point 1 to 2 is given by,

$$W_{1 \rightarrow 2} = F \cdot \Delta r$$

$$W_{1 \rightarrow 2} = F \Delta r \cos \theta$$

Since $\vec{F}$ and $\vec{\Delta r}$ are along the same direction therefore $\theta = 0$ and $\cos(0) = 1$,

$$W_{1 \rightarrow 2} = F \Delta r$$

By putting the values of Δr and F from equations (i) and (ii) we get,

$$W_{1 \rightarrow 2} = G \frac{m M_E}{r_1 r_2} (r_2 - r_1)$$

$$W_{1 \rightarrow 2} = G m M_E \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

$$W_{1 \rightarrow 2} = G m M_E \left(\frac{r_2}{r_1 r_2} - \frac{r_1}{r_1 r_2} \right)$$

$$W_{1 \rightarrow 2} = G m M_E \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Similarly, the work done in lifting the body from point 2 to 3, 3 to 4, ..., (n - 1) to n is given by,

$$W_{2 \rightarrow 3} = G m M_E \left(\frac{1}{r_2} - \frac{1}{r_3} \right)$$

$$W_{3 \rightarrow 4} = G m M_E \left(\frac{1}{r_3} - \frac{1}{r_4} \right)$$

$$W_{n-1 \rightarrow 1} = Gm M_E \left(\frac{1}{r_{n-1}} - \frac{1}{r_n} \right)$$

The total work done in lifting the body from point 1 to 'n' is given by,

$$W = W_{1 \rightarrow 2} + W_{2 \rightarrow 3} + W_{3 \rightarrow 4} + \dots + W_{n-1 \rightarrow n}$$

$$W = Gm M_E \left(\frac{1}{r_1} - \frac{1}{r_2} \right) + Gm M_E \left(\frac{1}{r_2} - \frac{1}{r_3} \right) + Gm M_E \left(\frac{1}{r_3} - \frac{1}{r_4} \right) + \dots + Gm M_E \left(\frac{1}{r_{n-1}} - \frac{1}{r_n} \right)$$

$$W = Gm M_E \left(\frac{1}{r_1} - \frac{1}{r_2} + \frac{1}{r_2} - \frac{1}{r_3} + \frac{1}{r_3} - \frac{1}{r_4} + \dots + \frac{1}{r_{n-1}} - \frac{1}{r_n} \right)$$

$$W = Gm M_E \left(\frac{1}{r_1} - \frac{1}{r_n} \right)$$

This work is stored in a body as potential energy, thus potential energy of the body at B with respect to A is given by,

$$P.E = Gm M_E \left(\frac{1}{r_1} - \frac{1}{r_n} \right)$$

The P.E of the body at point A with respect to point B is given by,

$$\Delta U = -W$$

$$\Delta U = -Gm M_E \left(\frac{1}{r_1} - \frac{1}{r_n} \right)$$

If the point B lies at infinity then $r_n = \infty$ then $1/\infty = 0$,

$$\Delta U = -Gm M_E \left(\frac{1}{r_1} \right)$$

This P.E of the body at point A is called **ABSOLUTE POTENTIAL ENERGY**

$$\Delta U = (P.E)_{Abs}$$

$$P.E_{Abs} = \frac{-Gm M_E}{r_1}$$

If R_E be the radius of earth then the absolute potential energy of the body at the surface of the earth is given by,

$$P.E_{Abs} = \frac{-Gm M_E}{R_E}$$

ABSOLUTE POTENTIAL ENERGY AT CERTAIN HEIGHT:

The absolute potential energy of the body at a height 'h' ($h \ll R_E$) above the surface of the earth is given by:

$$P.E_{Abs} = -\frac{Gm M_E}{R_E + h}$$

$$P.E_{Abs} = -\frac{Gm M_E}{R_E \left(1 + \frac{h}{R_E} \right)}$$

$$P.E_{Abs} = -\frac{Gm M_E}{R_E} \left(1 + \frac{h}{R_E} \right)^{-1}$$

Using binomial expansion,

$$P.E_{Abs} = -\frac{Gm M_E}{R_E} \left\{ 1 + \frac{(-1)}{1!} \frac{h}{R_E} + \frac{(-1)(-2)}{2!} \left(\frac{h}{R_E}\right)^2 + \dots \right\}$$

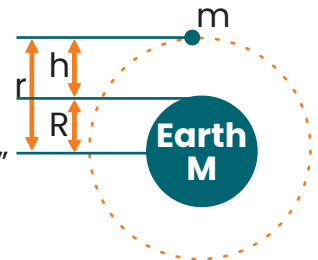
$$P.E_{Abs} = -\frac{Gm M_E}{R_E} \left\{ 1 - \frac{h}{R_E} + \left(\frac{h}{R_E}\right)^2 + \dots \right\}$$

Since $h \ll R_E$, we can neglect the terms containing the higher powers of h/R_E . Thus,

$$P.E_{Abs} = -\frac{Gm M_E}{R_E} \left(1 - \frac{h}{R_E}\right)_{-1}$$

ESCAPE VELOCITY:

"The minimum velocity with which the body has to be projected vertically upward from the surface of the earth or any other planet so that it just crosses the gravitational field of the planet and never returns."



EXPRESSION FOR ESCAPE VELOCITY:

Work is done at the cost of kinetic energy given to the body at the surface of the earth. If V is the escape velocity of the body projected from the surface of the earth, the kinetic energy of the body is equal to the gravitational potential energy

$$\frac{1}{2} m V^2 = \frac{Gm M}{r}$$

$$V^2 = \frac{2GM}{r}$$

$$V = \sqrt{\frac{2GM}{r}}$$

Where $g = \frac{GM}{r^2} \Rightarrow GM = gr^2$

$$V = \sqrt{\frac{2r^2}{r}}$$

$$V = \sqrt{2gr}$$

LAW OF CONSERVATION OF ENERGY:

Statement:

"Energy can neither be created nor be destroyed. It can only be transferred from one form to another form."

PROOF/ILLUSTRATION OR EXPLANATION:

A loss in one form of energy is accompanied by an equal increase in the other form of energy. The total energy remains constant.

Energy cannot be created means one cannot produce energy by expending nothing. We get energy only by expending something appropriate.

• Conservation of nuclear energy:

In nuclear reaction energy is created at the cost of mass. If 'm' is the mass annihilated then according to Einstein mass-energy relation, the energy produced is,

$$E = mc^2$$

We get something equivalent in return if we annihilated it. Pair production is an example of annihilation of energy.

• **Conservation of mechanical energy:**

Consider a body of mass 'm' placed at a point 'P' which is at height 'h' measured from the surface of the earth.

Potential and kinetic energy at point P,

Potential energy of the body at point P with respect to point 'O' lying on the surface of the earth equals to mgh.

$$P.E = mgh$$

$$K.E = 0$$

$$P.E + K.E = mgh$$

Potential and kinetic energy at point O,

$$P.E = mgh$$

$$P.E = mg(0)$$

$$P.E = 0$$

We now calculating kinetic energy at 'O' which is given by,

$$K.E = \frac{1}{2} mV^2$$

Where V is the velocity with which the body reaches the point 'O'. Using the equation of motion, we obtain the velocity as,

$$2as = V^2 - Vi^2$$

Where,

Initial velocity at point P $Vi = 0$

Final velocity at point O $V = v$

Substituting these in equation (ii) we get,

$$v^2 - 0 = 2gh$$

$$v^2 = 2gh$$

Substituting the value of V^2 in equation (i) we get,

$$K.E = \frac{1}{2} m(2gh)$$

$$K.E = mgh$$

The potential energy at 'O' is taken arbitrarily equal to zero with respect to which the potential energy at the point P is mgh.

$$P.E + K.E = mgh$$

We now calculate the potential and kinetic energy at any point 'Q' at a distance 'x' below the point P,

$$P.E = mg(h - x)$$

$$K.E = \frac{1}{2} mV^2$$

The velocity V is calculated in similar manner as before h is replaced by 'x',

$$K.E = mgx$$

$$\begin{aligned}
 \text{P.E} + \text{K.E} &= mg(h - x) + mgx \\
 \text{P.E} + \text{K.E} &= mgh - mgx + mgx \\
 \text{P.E} + \text{K.E} &= mgh
 \end{aligned}$$

Conclusion:

This shows that the sum of kinetic energy and potential energy (total energy) is always constant provided there is no force of friction involved during motion of the body.

Examples of interconversion of energy:

- When hands are rubbed the mechanical energy, due to friction changes to heat energy.
- When two stones are struck, the mechanical energy changes to heat and light energy.
- When knife is rubbed against grinding stone the mechanical energy changes to heat, light and sound energy.
- When brakes are applied the mechanical energy changes to heat energy at the point where brakes rub against the moving wheel.
- When a torch is switched on, the chemical energy of the cell changes to electric energy. The electrical energy on flowing through the filament of the bulb changes to heat and light.

WORK ENERGY EQUATION**OR****INTERCONVERSION OF KINETIC AND POTENTIAL ENERGY:**

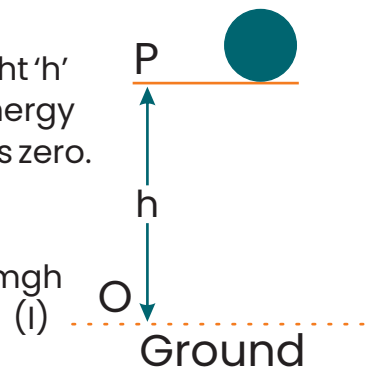
Consider a body of mass 'm' is placed at a point P which is at height 'h' from the ground. The body possesses a gravitational potential energy equal to 'mgh' with respect to the ground while its kinetic energy is zero.

$$\text{P.E of the body at P} = mgh$$

$$\text{K.E of the body at P} = 0$$

$$\text{Total energy of the body at P} = \text{K.E} + \text{P.E} = mgh$$

$$\text{Total energy at P} = mgh$$



If the body is allowed to fall freely under the action of gravity, then its potential energy will go on decreasing while its kinetic energy will go on increasing because the velocity of the body is increasing. If there is no force of friction acting on the body, then this P.E is converted to K.E of the body. When the body reaches just above the ground, its potential energy is nearly zero while its K.E is maximum, i.e. If V be the velocity of the body just before hitting the ground then K.E of the body is,

$$K.E = \frac{1}{2} mV^2$$

The velocity of the body can be found by formula,

$$2as = V^2 - Vi^2$$

Where,

Initial velocity at point P $Vi = 0$

Final velocity at point O $V = v$

Substituting these in equation (ii) we get,

$$V^2 - 0 = 2gh$$

$$V^2 = 2gh$$

$$K.E \text{ at } O = \frac{1}{2} m(2gh) = mgh$$

$$P.E \text{ at } O \text{ (near the ground)} = 0$$

$$\text{Total energy of the body at } O = K.E + P.E$$

$$\text{Total energy of the body at } O = mgh + 0$$

$$\text{Total energy of the body at } O = mgh \quad (ii)$$

From equation (i) and (ii),

$$\text{Loss of P.E} = \text{Gain in K.E}$$

In the presence of friction:

If there is some force of friction 'f' acting on the body then a fraction of its P.E is lost in doing work against the frictional force while remaining energy is converted into K.E.

$$\text{Loss of P.E} = \text{Gain of K.E} + \text{Work done against friction}$$

$$\text{Gain of K.E} = \text{Loss of P.E} - \text{Work done against friction}$$

$$\text{Loss of P.E} = mgh$$

$$\text{Work against friction} = f \cdot h$$

$$\text{Gain of K.E} = mgh - fh$$

$$\Delta K.E = mgh - fh$$

The above equation is termed as **Work – Energy equation**.